

Calc 3 – Vector Functions & Space Curves

Worksheet A

1. For $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$, find $\mathbf{r}'(t)$ and $\mathbf{r}''(t)$.
2. Find the unit tangent $\mathbf{T}(t)$ for $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ (helix).
3. Find the arc length of $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$ on $[0, 2\pi]$.
4. Compute the curvature of the circle $\mathbf{r}(t) = \langle 3 \cos t, 3 \sin t, 0 \rangle$.
5. For $\mathbf{r}(t) = \langle t, t^2, 0 \rangle$, compute the curvature at $t = 0$ using $\kappa = \frac{\|\mathbf{r}' \times \mathbf{r}''\|}{\|\mathbf{r}'\|^3}$.

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Worksheet B

1. For $\mathbf{r}(t) = \langle e^t, e^{-t}, t\sqrt{2} \rangle$, find $\|\mathbf{r}'(t)\|$.
2. Find the arc length of $\mathbf{r}(t) = \langle e^t, e^{-t}, t\sqrt{2} \rangle$ on $[0, 1]$.
3. For a particle with position $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$, find velocity, speed, and acceleration at $t = 1$.
4. Compute the tangential component of acceleration for $\mathbf{r}(t) = \langle t^2, t, 0 \rangle$ at $t = 1$.
5. For the plane curve $y = \frac{1}{x}$, $x > 0$, find $\kappa(x)$ using $\kappa = \frac{|y''|}{(1+(y')^2)^{3/2}}$, and evaluate at $x = 1$.

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Answer Key (Worksheets A & B)

Worksheet A.

1. $\mathbf{r}'(t) = \langle 1, 2t, 3t^2 \rangle$, $\mathbf{r}''(t) = \langle 0, 2, 6t \rangle$.

2. $\mathbf{r}'(t) = \langle -\sin t, \cos t, 1 \rangle$, $\|\mathbf{r}'(t)\| = \sqrt{2}$. $\mathbf{T} = \frac{1}{\sqrt{2}} \langle -\sin t, \cos t, 1 \rangle$.

3. $L = \int_0^{2\pi} \sqrt{2} dt = 2\sqrt{2}\pi$.

4. Circle of radius 3, so $\kappa = 1/3$ (constant).

5. $\mathbf{r}'(t) = \langle 1, 2t, 0 \rangle$, $\mathbf{r}''(t) = \langle 0, 2, 0 \rangle$. $\mathbf{r}' \times \mathbf{r}'' = \langle 0, 0, 2 \rangle$, $\|\mathbf{r}' \times \mathbf{r}''\| = 2$. $\|\mathbf{r}'\|^3 = (1 + 4t^2)^{3/2}$. At $t = 0$: $\kappa = 2$.

Worksheet B.

1. $\mathbf{r}'(t) = \langle e^t, -e^{-t}, \sqrt{2} \rangle$. $\|\mathbf{r}'\|^2 = e^{2t} + e^{-2t} + 2 = (e^t + e^{-t})^2$, so $\|\mathbf{r}'\| = e^t + e^{-t}$.

2. $L = \int_0^1 (e^t + e^{-t}) dt = [e^t - e^{-t}]_0^1 = e - 1/e$.

3. $\mathbf{v} = \langle 1, 2t, 3t^2 \rangle$, at $t = 1$: $\mathbf{v}(1) = \langle 1, 2, 3 \rangle$, speed = $\sqrt{14}$, $\mathbf{a}(1) = \langle 0, 2, 6 \rangle$.

4. $\mathbf{r}'(t) = \langle 2t, 1, 0 \rangle$, $\mathbf{r}''(t) = \langle 2, 0, 0 \rangle$. $a_T = \frac{\mathbf{r}' \cdot \mathbf{r}''}{\|\mathbf{r}'\|} = \frac{4t}{\sqrt{4t^2 + 1}}$. At $t = 1$: $4/\sqrt{5}$.

5. $y' = -1/x^2$, $y'' = 2/x^3$. $\kappa(x) = \frac{2/x^3}{(1 + 1/x^4)^{3/2}} = \frac{2x^3}{(x^4 + 1)^{3/2}}$. At $x = 1$: $\kappa = \frac{2}{2^{3/2}} = \frac{1}{\sqrt{2}}$.